

Estimation revisions on `codex/rulof_new_work`

Changes relative to the `efficient` branch

26 August 2026

1. Baseline transition model: Table 1 and Appendix Table A1

- Re-estimated the three-wave model using the exact eight-history likelihood.
- Repaired stationarity; used 12 reproducible starts (including nested-model starts); and required optimizer convergence, a small score, positive observed information, and valid likelihood nesting before saving results.
- Replaced stale inference with analytical survey-weighted sandwich and delta standard errors.
- Separated the free initial employment probability from the model-implied long-run employment share.

Table 1: Baseline latent-state model: implied probabilities (%)

	(1)	(2)	(3)	(4)	(5)	(6)
	No miscl.		Symmetric		Asymmetric	
	Stat.	Free	Stat.	Free	Stat.	Free
Entry rate (%)	8.18 (0.04)	8.28 (0.05)	2.86 (0.05)	2.98 (0.06)	2.86 (0.05)	2.98 (0.06)
Exit rate (%)	8.64 (0.04)	8.54 (0.05)	3.04 (0.05)	2.91 (0.06)	3.04 (0.05)	2.91 (0.06)
Misclassification rate (%)	—	—	2.99 (0.03)	2.99 (0.03)	—	—
False-positive rate (%)	—	—	—	—	3.05 (0.04)	3.05 (0.04)
False-negative rate (%)	—	—	—	—	2.93 (0.04)	2.92 (0.04)
Employment rate (%)	48.63 (0.09)	49.23 (0.18)	48.50 (0.09)	50.57 (0.53)	48.46 (0.09)	50.58 (0.53)
Log-likelihood	-357.208M	-357.204M	-349.378M	-349.371M	-349.376M	-349.368M
N	402,923	402,923	402,923	402,923	402,923	402,923

Note: Rates are percentages. Analytical SE in parentheses use an individual-level survey-weighted sandwich covariance matrix and the delta method. Employment is the model-implied steady-state share. The asymmetric specifications report false-positive and false-negative rates separately. The calculation does not incorporate strata or PSU clustering.

Appendix Table A1: Baseline latent-state model: parameter estimates

	(1)	(2)	(3)	(4)	(5)	(6)
	No miscl.		Symmetric		Asymmetric	
	Stat.	Free	Stat.	Free	Stat.	Free
<i>Transition parameters</i>						
θ_0 (entry)	0.0818 (0.0004)	0.0828 (0.0005)	0.0286 (0.0005)	0.0298 (0.0006)	0.0286 (0.0005)	0.0298 (0.0006)
θ_1 (persistence)	0.9136 (0.0004)	0.9146 (0.0005)	0.9696 (0.0005)	0.9709 (0.0006)	0.9696 (0.0005)	0.9709 (0.0006)
α (initial empl.)	0.4863 (0.0009)	0.4852 (0.0009)	0.4850 (0.0009)	0.4837 (0.0010)	0.4846 (0.0009)	0.4832 (0.0010)
<i>Misclassification</i>						
π (symmetric)	—	—	0.0299 (0.0003)	0.0299 (0.0003)	—	—
π_0 (false positive)	—	—	—	—	0.0305 (0.0004)	0.0305 (0.0004)
π_1 (false negative)	—	—	—	—	0.0293 (0.0004)	0.0292 (0.0004)
Log-likelihood	-357.208M	-357.204M	-349.378M	-349.371M	-349.376M	-349.368M
N	402,923	402,923	402,923	402,923	402,923	402,923

Note: Analytical SE in parentheses use an individual-level survey-weighted sandwich covariance matrix. α is the initial employment probability; in stationary columns it is derived from the transition parameters. The calculation does not incorporate strata or PSU clustering.

Interpretation

- The no-error model implies quarterly entry and exit of about 8.2–8.6%.
- Allowing symmetric error lowers both rates to about 3%, with a per-wave misclassification probability of 2.99%.
- False-positive and false-negative probabilities are very similar, so asymmetry adds little to the substantive result.
- The repaired likelihood strengthens the paper’s central conclusion: relatively infrequent classification error can create a large upward bias in measured labour-market turnover.

To do

- Report likelihood-ratio evidence for stationarity and asymmetric error, and check whether transition probabilities vary across wave pairs.
- Confirm the analytical inference with a person-level bootstrap or likelihood profiles for the misclassification and transition rates.

2. Panel-matching robustness: Table 2

- Applied the same exact, non-stationary estimator, 12-start search, and convergence and curvature checks to panels A, B, and C.
- Held all sample filters fixed apart from progressively stricter matching.
- Added consistent analytical standard errors to every panel comparison.

Table 2: Sensitivity to alternative panel-matching rules

	Baseline (A)		Stricter (B)		Strictest (C)	
	No error	Symmetric error	No error	Symmetric error	No error	Symmetric error
Entry rate (%)	8.28 (0.05)	2.98 (0.06)	7.59 (0.05)	2.81 (0.06)	6.56 (0.05)	2.63 (0.06)
Exit rate (%)	8.54 (0.05)	2.91 (0.06)	7.55 (0.05)	2.48 (0.06)	6.03 (0.05)	1.93 (0.06)
Misclassification rate (%)	—	2.99 (0.03)	—	2.67 (0.03)	—	2.15 (0.03)
Initial employment rate (%)	48.52 (0.09)	48.37 (0.10)	48.52 (0.09)	48.40 (0.10)	48.94 (0.11)	48.85 (0.11)
Employment rate (%)	49.23 (0.18)	50.57 (0.53)	50.11 (0.20)	53.10 (0.58)	52.13 (0.25)	57.64 (0.70)
Log-likelihood (millions)	-357.204	-349.371	-324.427	-317.258	-243.918	-238.714
N	402,923	402,923	377,132	377,132	300,200	300,200

Note: All models estimate the initial employment probability freely; stationarity is not imposed. Long-run employment is $\theta_0/(\theta_0 + 1 - \theta_1)$. Parentheses contain individual-level survey-weighted sandwich/delta standard errors. Panels A, B, and C apply successively stricter matching rules and use otherwise identical sample filters.

Interpretation

- Stricter matching lowers both observed and corrected transition rates.
- The estimated misclassification probability declines from 2.99% in panel A to 2.15% in panel C.
- Corrected entry and exit remain much lower than no-error rates under every matching rule.
- Matching quality is therefore part of the inconsistency mechanism, but the main correction is not driven solely by the baseline matching rule.

To do

- Compare observable composition and attrition across panels A–C, not only their parameter estimates.
- Check whether the Table 4 and age/education-consistency results are also stable under the stricter matching rules.

3. Age and education consistency: Table 3 and Appendix Table A2

- Re-estimated both the indicator-only model and the model with additional effects for large age and education increments, under stationarity and with a free initial employment probability.
- Repaired the EM update, used multiple starts and likelihood checks, and added analytical standard errors.
- Interpreted age and education inconsistencies neutrally: they may arise from response error, matching error, or both.
- Added robustness allowing true transition rates to differ between records with and without an inconsistency and allowing misclassification to depend on inconsistency severity.
- Added stationary and free-initial-employment specifications in which the misclassification rate also depends on person-wave indicators for membership in matching panel B but not C and panel A but not B, defined using the Table 2 sample-construction rules.

Table 3: Age- and education-inconsistency models

	(1)	(2)	(3)	(4)	(5)	(6)
	Indicators only		Indicators + large increments		Plus matching indicators	
	Stationary	Free α	Stationary	Free α	Stationary	Free α
Entry rate (%)	2.32**	2.60**	2.31**	2.60**	2.28**	2.57**
	(0.05)	(0.05)	(0.05)	(0.05)	(0.04)	(0.05)
Exit rate (%)	2.52**	2.20**	2.52**	2.19**	2.47**	2.14**
	(0.05)	(0.06)	(0.05)	(0.06)	(0.05)	(0.06)
Misclassification rate (%)	3.35**	3.35**	3.36**	3.36**	3.38**	3.38**
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Initial employment rate (%)	47.92**	47.60**	47.92**	47.60**	47.92**	47.59**
	(0.09)	(0.10)	(0.09)	(0.10)	(0.09)	(0.10)
Log-likelihood	-333.651M	-333.601M	-333.352M	-333.301M	-332.881M	-332.826M
N	394,823	394,823	394,823	394,823	394,823	394,823
Stationarity imposed	Yes	No	Yes	No	Yes	No
Large-increment effects	No	No	Yes	Yes	Yes	Yes
Matching-quality indicators	No	No	No	No	Yes	Yes

Note: Entry, exit, and mean misclassification probabilities are reported on a common percentage scale. Mean misclassification averages individual-wave predictions using survey weights. Columns (5)–(6) add person-wave indicators for observations included in matching panel B but not C and in panel A but not B, using the Table 2 sample-construction rules. Analytical standard errors in parentheses use the survey-weighted sandwich covariance and delta method. Under stationarity, initial and steady-state employment coincide. An inconsistency may arise from response error, matching error, or both.

Appendix Table A2: Age- and education-inconsistency models: supplementary estimates

	(1)	(2)	(3)	(4)	(5)	(6)
	Indicators only		Indicators + large increments		Plus matching indicators	
	Stationary	Free α	Stationary	Free α	Stationary	Free α
<i>Employment-state transformations</i>						
Steady-state employment (%)	47.92** (0.09)	54.17** (0.62)	47.92** (0.09)	54.28** (0.62)	47.92** (0.09)	54.57** (0.63)
Initial minus steady employment (p.p.)	0.00 (fixed)	-6.56** (0.64)	0.00 (fixed)	-6.68** (0.65)	0.00 (fixed)	-6.98** (0.65)
<i>Misclassification by inconsistency pattern</i>						
No inconsistency (%)	2.25** (0.03)	2.26** (0.03)	2.27** (0.03)	2.27** (0.03)	1.98** (0.03)	1.98** (0.03)
Age inconsistency (%)	17.96** (0.26)	17.99** (0.26)	7.76** (0.37)	7.76** (0.36)	5.46** (0.28)	5.46** (0.28)
Age inconsistency, large (%)	17.96** (0.26)	17.99** (0.26)	21.46** (0.33)	21.50** (0.33)	15.83** (0.33)	15.86** (0.33)
Education inconsistency (%)	6.48** (0.12)	6.49** (0.12)	5.99** (0.18)	5.99** (0.18)	4.47** (0.15)	4.47** (0.15)
Education inconsistency, large (%)	6.48** (0.12)	6.49** (0.12)	6.58** (0.15)	6.60** (0.15)	4.82** (0.14)	4.83** (0.14)
Both inconsistencies (%)	31.94** (0.30)	31.99** (0.30)	17.24** (0.71)	17.24** (0.70)	11.31** (0.59)	11.30** (0.59)
Both inconsistencies, both large (%)	31.94** (0.30)	31.99** (0.30)	35.28** (0.35)	35.34** (0.35)	27.28** (0.50)	27.33** (0.50)
Panel B but not C, no inconsistency (%)	—	—	—	—	2.77** (0.06)	2.77** (0.06)
Panel A but not B, no inconsistency (%)	—	—	—	—	4.33** (0.09)	4.34** (0.09)
<i>Probability increments</i>						
Additional large age increment (p.p.)	0.00 (fixed)	0.00 (fixed)	13.70** (0.48)	13.73** (0.48)	10.37** (0.39)	10.40** (0.39)
Additional large education increment (p.p.)	0.00 (fixed)	0.00 (fixed)	0.59** (0.23)	0.60** (0.23)	0.35* (0.17)	0.36* (0.17)
Panel B-not-C increment (p.p.)	—	—	—	—	0.79** (0.05)	0.79** (0.05)
Panel A-not-B increment (p.p.)	—	—	—	—	2.35** (0.08)	2.36** (0.08)
<i>Logistic-link coefficients</i>						
δ_0 (intercept)	-3.0530** (0.0145)	-3.0526** (0.0145)	-3.0467** (0.0144)	-3.0463** (0.0144)	-3.1893** (0.0162)	-3.1888** (0.0161)
δ_1 (age inconsistency)	2.4742** (0.0247)	2.4766** (0.0247)	1.3525** (0.0562)	1.3520** (0.0561)	1.0904** (0.0558)	1.0891** (0.0557)
δ_2 (education inconsistency)	1.1487** (0.0220)	1.1502** (0.0220)	1.0521** (0.0340)	1.0523** (0.0339)	0.8695** (0.0349)	0.8690** (0.0348)
δ_3 (large age increment)	0.00 (fixed)	0.00 (fixed)	1.4089** (0.0610)	1.4120** (0.0609)	1.3297** (0.0597)	1.3329** (0.0596)
δ_4 (large education increment)	0.00 (fixed)	0.00 (fixed)	0.1077** (0.0414)	0.1099** (0.0413)	0.0826* (0.0401)	0.0846* (0.0400)
δ_5 (panel B but not C)	0.000 (fixed)	0.000 (fixed)	0.000 (fixed)	0.000 (fixed)	0.3514** (0.0215)	0.3516** (0.0215)
δ_6 (panel A but not B)	0.000 (fixed)	0.000 (fixed)	0.000 (fixed)	0.000 (fixed)	0.8339** (0.0239)	0.8358** (0.0239)
Log-likelihood	-333.651M	-333.601M	-333.352M	-333.301M	-332.881M	-332.826M
N	394,823	394,823	394,823	394,823	394,823	394,823
Stationarity imposed	Yes	No	Yes	No	Yes	No
Large-increment effects	No	No	Yes	Yes	Yes	Yes
Matching-quality indicators	No	No	No	No	Yes	Yes

Note: Probabilities and increments are percentages or percentage points; logit coefficients are untransformed. A large inconsistency lies at least two units beyond the admissible progression. Panel-membership probabilities set age and education inconsistencies to zero and vary one matching indicator at a time. Analytical standard errors use the survey-weighted sandwich covariance and delta method.

The additional reliability-group robustness table is generated as table_inconsistency_reliability.tex in EM-baseline-ext/output/tables.

Interpretation

- Baseline misclassification is 2.26%, rising to 17.99% with an age inconsistency, 6.49% with an education inconsistency, and 31.99% with both.
- Stationarity materially changes the entry and exit decomposition but barely changes the inconsistency-related misclassification probabilities. In the free model, initial employment is 6.56 percentage points below its implied steady-state rate.
- Allowing additional large-increment effects substantially sharpens the age gradient: predicted misclassification rises from about 7.76% for a mild age inconsistency to 21.50% for a large one. The corresponding education increase is only about 0.60 percentage points.
- In the augmented free-initial-employment model, predicted misclassification for a record with no age or education inconsistency is 1.98% when retained by both stricter panels, 2.77% when retained by B but not C, and 4.34% when retained by A but not B. Thus the matching-quality indicators capture a strong reliability gradient.
- Controlling for matching quality attenuates but does not eliminate the age/education gradients: predicted error is 5.46% for a mild and 15.86% for a large age inconsistency, and 4.47% and 4.83% for mild and large education inconsistencies. The mean error rate changes little, from 3.36% to 3.38%, and the free-model entry and exit rates move only from 2.60% and 2.19% to 2.57% and 2.14%.
- The relationship is not explained solely by different true transition rates among records with age or education inconsistencies.
- Larger age inconsistencies are much more informative than mild ones; inconsistency severity matters less for education.
- Age and education inconsistencies predict higher status misclassification, without distinguishing response errors from incorrect panel matches.

To do

- Check alternative nonlinear definitions of inconsistency extent and more flexible inconsistency-group transition rates.
- Bootstrap the age/education-inconsistency effects and repeat the analysis under stricter panel matching.

4. Observable transition heterogeneity: Table 4

- Repaired the covariate-model optimiser so every specification maximises the appropriate likelihood, uses five starts including a nested-model warm start, and dominates its nested no-error model.
- Added analytical standard errors and posterior risk-set and survey-weighted transition rates.
- Added tenure and time since work in Set 3.
- Added time-varying contract type and informal-sector status in Set 4 without restricting the sample to respondents observed in employment at every wave.
- Added a Set 4 specification in which the symmetric misclassification rate depends on age and education inconsistencies, their severity, and the two nested panel-matching indicators used in Table 3, columns (5)–(6).
- Moved raw coefficients, the initial-state parameter, total churn, and hazard distributions to an appendix table.

Table 4: Risk-set and survey-weighted quarterly employment transitions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	No miscl.				Constant symmetric				Reliability-dependent
	Set 1	Set 2	Set 3	Set 4	Set 1	Set 2	Set 3	Set 4	Set 4
Entry rate (%)	8.14 (0.05)	8.14 (0.05)	8.14 (0.05)	8.14 (0.05)	3.62 (0.06)	3.77 (0.06)	4.66 (0.06)	4.87 (0.06)	4.67 (0.06)
Exit rate (%)	8.58 (0.05)	8.58 (0.05)	8.58 (0.05)	8.58 (0.05)	3.71 (0.06)	3.89 (0.06)	4.81 (0.06)	5.09 (0.06)	4.62 (0.06)
Misclassification rate (%)	—	—	—	—	2.58 (0.03)	2.50 (0.03)	2.34 (0.04)	2.18 (0.03)	2.55 (0.03)
Employment rate (%)	47.94 (0.09)	47.94 (0.09)	47.94 (0.09)	47.94 (0.09)	47.83 (0.09)	47.84 (0.09)	47.85 (0.09)	47.87 (0.09)	47.87 (0.01)
Log-likelihood	-343.8M	-342.6M	-337.1M	-336.2M	-338.2M	-337.5M	-333.2M	-332.7M	-324.0M
<i>N</i>	394,823	394,823	394,823	394,823	394,823	394,823	394,823	394,823	394,823
<i>Covariates included</i>									
Age, Age ² , Educ.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
+ Race, Gender	No	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
+ Tenure (exit), time since work (entry)	No	No	Yes	Yes	No	No	Yes	Yes	Yes
+ Contract type, informal sector (exit only)	No	No	No	Yes	No	No	No	Yes	Yes
Reliability variables in error equation	No	No	No	No	No	No	No	No	Yes

Note: All specifications estimate the initial employment probability freely. Rates average predicted hazards using survey weights and posterior probabilities of belonging to the relevant origin-state risk set. Employment is the survey-weighted posterior employed share. SE are bootstrap estimates when available and otherwise use an individual-level survey-weighted sandwich/delta approximation. The approximation does not incorporate strata or PSU clustering. Column (9) retains the Set 4 transition equations and lets the symmetric error probability depend on age and education inconsistencies, their severity, and the two nested matching-quality indicators used in Table 3, columns (5)–(6).

Appendix Table A3: Covariate-model coefficients and predicted-hazard distributions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	No miscl.				Symmetric			
	Set 1	Set 2	Set 3	Set 4	Set 1	Set 2	Set 3	Set 4
<i>Other derived model quantities</i>								
Initial employment rate (%)	47.93** (0.09)	47.93** (0.09)	47.93** (0.09)	47.93** (0.09)	47.80** (0.10)	47.82** (0.10)	47.84** (0.10)	47.87** (0.10)
Total employment churn (%)	8.35** (0.04)	8.35** (0.04)	8.35** (0.04)	8.35** (0.04)	3.66** (0.05)	3.83** (0.05)	4.73** (0.05)	4.98** (0.05)
<i>Entry equation: $\Phi(X_{it}\beta_0)$</i>								
Intercept	-1.3658** (0.0034)	-1.2464** (0.0053)	-0.8703** (0.0069)	-0.8707** (0.0069)	-1.7943** (0.0077)	-1.6189** (0.0093)	-1.0588** (0.0100)	-1.0416** (0.0097)
Age	1.3871** (0.0239)	1.4214** (0.0241)	1.3336** (0.0266)	1.3364** (0.0266)	1.9100** (0.0457)	1.9325** (0.0454)	1.4236** (0.0526)	1.4300** (0.0490)
Age ²	-1.2894** (0.0247)	-1.3104** (0.0249)	-1.2065** (0.0264)	-1.2092** (0.0264)	-1.8170** (0.0486)	-1.8232** (0.0481)	-1.3515** (0.0512)	-1.3505** (0.0480)
Education	0.0283** (0.0037)	0.0386** (0.0038)	0.0071 (0.0039)	0.0071 (0.0039)	0.0313** (0.0065)	0.0483** (0.0068)	-0.0186** (0.0065)	-0.0137* (0.0063)
Race: Coloured	—	0.1472** (0.0111)	0.0632** (0.0113)	0.0631** (0.0113)	—	0.1508** (0.0190)	0.0720** (0.0174)	0.0715** (0.0169)
Race: Indian	—	-0.1557** (0.0263)	-0.1097** (0.0269)	-0.1096** (0.0269)	—	-0.2941** (0.0571)	-0.2356** (0.0532)	-0.2390** (0.0514)
Race: White	—	-0.0678** (0.0207)	-0.0535* (0.0211)	-0.0532* (0.0211)	—	-0.1598** (0.0409)	-0.1423** (0.0378)	-0.1506** (0.0368)
Female	—	-0.2247** (0.0068)	-0.1600** (0.0069)	-0.1599** (0.0069)	—	-0.2948** (0.0120)	-0.2044** (0.0115)	-0.2015** (0.0111)
log(1 + tenure)	—	—	0.0000 [fixed]	0.0000 [fixed]	—	—	0.0000 [fixed]	0.0000 [fixed]
log(1 + timesincework)	—	—	-0.2455** (0.0043)	-0.2455** (0.0043)	—	—	-0.3101** (0.0066)	-0.3071** (0.0064)
Never worked	—	—	-0.5417** (0.0086)	-0.5413** (0.0086)	—	—	-0.8704** (0.0241)	-0.8352** (0.0212)
Tenure missing	—	—	0.0000 [fixed]	0.0000 [fixed]	—	—	0.0000 [fixed]	0.0000 [fixed]
Permanent contract	—	—	—	0.0000 [fixed]	—	—	—	0.0000 [fixed]
Informal sector	—	—	—	0.0000 [fixed]	—	—	—	0.0000 [fixed]
<i>Employment-persistence equation: $\Phi(X_{it}\beta_1)$</i>								
Intercept	1.2840** (0.0037)	1.2772** (0.0053)	1.1032** (0.0058)	1.0437** (0.0085)	1.7297** (0.0079)	1.6890** (0.0103)	1.3743** (0.0098)	1.2854** (0.0130)
Age	0.8081** (0.0305)	0.9015** (0.0309)	0.6830** (0.0317)	0.6915** (0.0318)	0.9079** (0.0550)	1.0118** (0.0543)	0.6174** (0.0529)	0.6434** (0.0505)
Age ²	-0.5637** (0.0295)	-0.6602** (0.0298)	-0.5788** (0.0306)	-0.5799** (0.0307)	-0.5689** (0.0546)	-0.6765** (0.0536)	-0.4860** (0.0526)	-0.5117** (0.0499)
Education	0.2019** (0.0032)	0.1756** (0.0034)	0.1438** (0.0036)	0.1049** (0.0038)	0.2461** (0.0048)	0.2118** (0.0051)	0.1720** (0.0057)	0.1230** (0.0058)
Race: Coloured	—	0.0969** (0.0104)	0.0800** (0.0105)	0.0197 (0.0107)	—	0.1348** (0.0191)	0.0510** (0.0172)	-0.0168 (0.0168)
Race: Indian	—	0.3125** (0.0250)	0.2556** (0.0259)	0.2307** (0.0262)	—	0.6983** (0.0861)	0.5734** (0.0647)	0.5341** (0.0626)
Race: White	—	0.3926** (0.0168)	0.3432** (0.0174)	0.3083** (0.0175)	—	0.8114** (0.0669)	0.7096** (0.0518)	0.6342** (0.0470)
Female	—	-0.0973** (0.0070)	-0.0744** (0.0071)	-0.0845** (0.0072)	—	-0.1181** (0.0123)	-0.0548** (0.0119)	-0.0678** (0.0115)
log(1 + tenure)	—	—	0.3214** (0.0048)	0.2998** (0.0049)	—	—	0.5980** (0.0151)	0.5083** (0.0120)
log(1 + timesincework)	—	—	0.0000 [fixed]	0.0000 [fixed]	—	—	0.0000 [fixed]	0.0000 [fixed]
Never worked	—	—	0.0000 [fixed]	0.0000 [fixed]	—	—	0.0000 [fixed]	0.0000 [fixed]
Tenure missing	—	—	-0.0350 (1.0497)	-0.0398 (1.0311)	—	—	-0.0938 (1.2670)	-0.0993 (1.2935)
Permanent contract	—	—	—	0.2191** (0.0086)	—	—	—	0.2657** (0.0135)
Informal sector	—	—	—	-0.1943** (0.0094)	—	—	—	-0.2417** (0.0136)
<i>Distribution of predicted entry hazards</i>								
10th percentile	3.86	3.67	3.21	3.20	1.07	0.97	0.89	1.00
Median	8.17	7.86	5.99	5.99	3.37	3.29	2.00	2.21
90th percentile	12.41	13.59	17.45	17.44	6.54	7.62	13.46	13.82
<i>Distribution of predicted exit hazards</i>								
10th percentile	4.37	3.29	2.15	1.94	1.02	0.41	0.14	0.19
Median	7.56	7.69	7.14	6.78	2.75	3.00	2.28	2.51
90th percentile	14.56	14.84	17.09	18.05	7.64	8.40	13.53	14.29
<i>Set 4 discrete effects on the exit probability</i>								
Permanent contract	—	—	—	-3.28** (0.13)	—	—	—	-2.43** (0.13)
Informal sector	—	—	—	3.01** (0.16)	—	—	—	2.36** (0.15)

Note: Raw probit coefficients are reported. Tenure enters persistence only; time since work and never worked enter entry only. Contract type and sector are origin-wave characteristics and may change between transitions; their entry coefficients are fixed at zero because these attributes are not defined for non-employed origin states. Hazard percentiles use survey and posterior risk-set weights. SE are bootstrap estimates when available and otherwise use an individual-level survey-weighted sandwich/delta approximation; quantile SE are omitted.

Appendix Table A3B: Reliability-dependent misclassification estimates for Table 4

Set 4; reliability-dependent error	
<i>Symmetric-error equation: $0.5 \Lambda(Z_{it}\delta)$</i>	
Intercept	-3.7428 (0.0244)
Age inconsistency	1.4368 (0.0691)
Education inconsistency	1.0714 (0.0490)
Large age inconsistency	1.5202 (0.0717)
Large education inconsistency	0.2802 (0.0536)
Panel B but not C	0.3391 (0.0312)
Panel A but not B	1.0175 (0.0319)
<i>Implied misclassification probabilities</i>	
No inconsistency; panel C (%)	1.16 (0.03)
Age inconsistency (%)	4.53 (0.29)
Large age inconsistency (%)	15.65 (0.41)
Education inconsistency (%)	3.23 (0.16)
Large education inconsistency (%)	4.19 (0.16)
Both inconsistencies (%)	11.27 (0.75)
Both large inconsistencies (%)	31.89 (0.66)
Panel B but not C (%)	1.61 (0.05)
Panel A but not B (%)	3.07 (0.08)

Note: Parentheses contain analytical individual-level survey-weighted sandwich/delta-method standard errors. Each implied rate changes the named indicator(s) from the panel-C, internally consistent reference case while holding the remaining indicators at zero.

Interpretation

- No-error entry and exit remain close to 8.1% and 8.6% across covariate sets.
- Symmetric-error estimates range from 3.62–4.87% for entry and 3.71–5.09% for exit; estimated misclassification falls from 2.58% in Set 1 to 2.18% in Set 4.
- In symmetric Set 4, predicted risks are highly dispersed: the weighted 10th–90th percentile range is 1.00–13.82% for entry and 0.19–14.29% for exit.
- Allowing error rates to depend on linked-record reliability gives entry and exit rates of 4.67% and 4.62%, and raises the mean error rate from 2.18% in constant-error Set 4 to 2.55%.
- The implied error probability is only 1.16% for an internally consistent panel-C record, but rises to 3.07% for records in panel A but not B, 15.65% for a large age inconsistency, and 31.89% when both age and education have large inconsistencies. Reliability heterogeneity is therefore substantial, although corrected entry and exit remain close to the original Set 4

values.

- Observable heterogeneity therefore explains some differences in transition risk but does not explain away the classification-error signal.
- Duration is economically important: entry falls with time since work and exit falls with tenure.
- Permanent contracts reduce exit and informal employment raises it. These are conditional associations, not causal estimates.
- The paper should distinguish heterogeneous true transition risk from the upward bias in observed transitions caused by classification error.

To do

- Allow the initial employment probability to depend on baseline covariates and check whether this changes the transition and error estimates.
- Test more flexible duration functions and confirm the main inference with a person-level bootstrap.
- Check robustness to alternative definitions and interactions for the reliability indicators in the misclassification equation.

5. Higher-order state dependence: Table 5

- Added the comparable four-wave AR(1) benchmark with symmetric error and free initial employment.
- Added the same tenure and time-since-work controls used in Table 4.
- Allowed duration covariates to change across origin waves.
- Added a nested AR(2) extension with Set 4 transition controls, followed by inconsistency/severity and matching-quality predictors of symmetric error.
- Used informed and perturbed starts, exact stage-specific likelihood collapse, likelihood-nesting checks, Newton polishing, and score/rank diagnostics.
- Replaced approximate optimisation and inference with exact collapsed likelihood estimation and analytical standard errors.

Table 5: Four-wave AR(1) and AR(2) estimates

	AR(1)	AR(2)	AR(2), duration controls	AR(2), Set 4	+ inconsistency	+ matching quality
Entry rate: nonemployed in both preceding waves	3.20 (0.04)	4.60 (0.06)	4.51 (0.07)	4.72 (0.06)	4.22 (0.05)	4.22 (0.05)
Entry rate: employed two waves ago	3.20 (0.04)	30.75 (0.47)	17.48 (0.97)	31.13 (0.53)	27.52 (0.78)	28.28 (0.71)
Exit rate: nonemployed two waves ago	3.13 (0.04)	31.60 (0.48)	19.16 (0.81)	32.37 (0.50)	28.65 (0.68)	29.38 (0.62)
Exit rate: employed in both preceding waves	3.13 (0.04)	4.38 (0.06)	4.27 (0.08)	4.56 (0.06)	3.95 (0.05)	3.94 (0.05)
Misclassification rate	2.70 (0.03)	0.98 (0.04)	1.49 (0.08)	0.93 (0.05)	2.02 (0.05)	1.95 (0.04)
Initial employment rate	47.97 (0.11)	48.11 (0.10)	48.10 (0.11)	47.83 (0.11)	47.94 (0.11)	47.93 (0.11)
Log likelihood	-468,443.8	-467,392.7	-462,205.1	-445,472.5	-434,650.5	-434,035.7
N	317,437	317,437	317,437	308,240	308,240	308,240
Parameters	4	8	13	29	33	35
Second employment lag	No	Yes	Yes	Yes	Yes	Yes
Duration controls	No	No	Yes	Yes	Yes	Yes
Set 4 transition controls	No	No	No	Yes	Yes	Yes
Error predictors	Constant	Constant	Constant	Constant	Inconsistency/severity	+ matching quality

Notes. Entries are model-implied quarterly probabilities in percent. Covariate specifications report posterior risk-set and survey-weighted averages; the misclassification row is the corresponding posterior mean probability. Initial employment is the latent employment probability in wave 2. Parentheses contain analytical sandwich/delta-method standard errors. The AR(1) rate is repeated across the two corresponding AR(2) rows because the first-order model imposes equality across the second-lag state. Every specification has one latent type, symmetric misclassification, and free initial conditions. Set 4 includes the Table 4 demographic, tenure, time-since-work, contract, and informal-sector transition controls. Inconsistency/severity adds age and education inconsistency indicators and large-inconsistency indicators to the error equation; matching quality additionally adds panel B-not-C and panel A-not-B indicators. The three extensions use the same 308,240 observations and exact nested likelihoods.

Appendix Table A4: Four-wave AR(1) and AR(2) coefficients

Coefficient	AR(1)	AR(2)	AR(2), duration controls
Initial employment logit	-0.081 (0.004)	—	—
Initial-pair logit: 10	—	-2.677 (0.016)	-2.847 (0.031)
Initial-pair logit: 01	—	-2.674 (0.016)	-2.839 (0.030)
Initial-pair logit: 11	—	-0.081 (0.004)	-0.081 (0.004)
Entry: intercept	-1.852 (0.006)	-1.685 (0.006)	-1.207 (0.012)
Entry: employed two waves ago	—	1.182 (0.014)	0.811 (0.034)
Entry: log time since work	—	—	-0.215 (0.010)
Entry: never worked	—	—	-0.831 (0.028)
Entry: timegap missing	—	—	-0.132 (0.082)
Persistence: intercept	1.862 (0.006)	0.479 (0.013)	0.575 (0.025)
Persistence: employed two waves ago	—	1.230 (0.014)	0.903 (0.025)
Persistence: log tenure	—	—	0.431 (0.015)
Persistence: tenure missing	—	—	0.003 (0.719)
Misclassification logit	-2.862 (0.010)	-3.917 (0.042)	-3.486 (0.055)

Appendix Table A4B: Set 4 AR(2) extension coefficients

Coefficient	Set 4	+ inconsistency	+ matching quality
Initial-pair logit: 10	-2.696 (0.019)	-3.148 (0.029)	-3.102 (0.026)
Initial-pair logit: 01	-2.685 (0.019)	-3.009 (0.025)	-2.972 (0.023)
Initial-pair logit: 11	-0.094 (0.004)	-0.093 (0.004)	-0.093 (0.004)
Entry: intercept	-1.154 (0.011)	-1.110 (0.012)	-1.117 (0.012)
Entry: employed two waves ago	0.860 (0.019)	0.665 (0.032)	0.693 (0.028)
Entry: age	1.113 (0.038)	1.126 (0.043)	1.124 (0.043)
Entry: age squared	-1.045 (0.037)	-1.059 (0.043)	-1.054 (0.042)
Entry: education	0.005 (0.005)	0.001 (0.006)	0.001 (0.006)
Entry: race 2	0.031 (0.015)	0.026 (0.017)	0.017 (0.017)
Entry: race 3	-0.151 (0.041)	-0.205 (0.047)	-0.222 (0.047)
Entry: race 4	-0.105 (0.032)	-0.138 (0.037)	-0.136 (0.037)
Entry: female	-0.143 (0.010)	-0.142 (0.011)	-0.140 (0.011)
Entry: log time since work	-0.215 (0.007)	-0.300 (0.009)	-0.293 (0.008)
Entry: never worked	-0.585 (0.015)	-0.772 (0.022)	-0.755 (0.020)
Entry: timegap missing	-0.332 (0.201)	-0.455 (0.212)	-0.440 (0.210)
Persistence: intercept	0.471 (0.018)	0.603 (0.026)	0.582 (0.024)
Persistence: employed two waves ago	0.882 (0.017)	0.712 (0.027)	0.744 (0.024)
Persistence: age	0.567 (0.044)	0.533 (0.050)	0.538 (0.050)
Persistence: age squared	-0.468 (0.043)	-0.459 (0.050)	-0.463 (0.049)
Persistence: education	0.101 (0.005)	0.085 (0.006)	0.087 (0.006)
Persistence: race 2	-0.010 (0.015)	-0.047 (0.017)	-0.040 (0.017)
Persistence: race 3	0.270 (0.041)	0.319 (0.048)	0.346 (0.049)
Persistence: race 4	0.381 (0.032)	0.428 (0.038)	0.432 (0.038)
Persistence: female	-0.062 (0.010)	-0.061 (0.012)	-0.064 (0.011)
Persistence: log tenure	0.299 (0.009)	0.484 (0.017)	0.461 (0.015)
Persistence: tenure missing	0.390 (0.638)	1.090 (0.500)	1.585 (0.405)
Persistence: permanent contract	0.196 (0.012)	0.322 (0.015)	0.310 (0.014)
Persistence: informal sector	-0.168 (0.013)	-0.143 (0.015)	-0.151 (0.015)
Error: intercept	-3.966 (0.052)	-3.938 (0.046)	-4.251 (0.054)
Error: age inconsistency	—	2.413 (0.052)	2.090 (0.052)
Error: education inconsistency	—	1.497 (0.043)	1.298 (0.046)
Error: large age inconsistency	—	1.063 (0.054)	0.966 (0.053)

Coefficient	Set 4	+ inconsistency	+ matching quality
Error: large education inconsistency	—	0.166 (0.056)	0.160 (0.054)
Error: panel B not C	—	—	0.447 (0.043)
Error: panel A not B	—	—	1.247 (0.048)

Appendix Table A4C: Implications for apparent two-wave transitions

Model	Classification only	Genuine reversal	Genuine persistence
<i>All apparent transitions</i>			
Table 1: raw/no error	0.0	8.4	91.6
Table 1: symmetric-error adjustment	67.0	1.0	32.1
Table 5: final AR(2)	41.1	16.8	42.1
<i>Apparent entries</i>			
Table 1: raw/no error	0.0	8.5	91.5
Table 1: symmetric-error adjustment	66.0	1.0	33.0
Table 5: final AR(2)	38.9	17.7	43.4
<i>Apparent exits</i>			
Table 1: raw/no error	0.0	8.3	91.7
Table 1: symmetric-error adjustment	67.9	1.0	31.1
Table 5: final AR(2)	43.4	15.9	40.8

Notes. Entries are survey-weighted percentages conditional only on the two reports forming an apparent transition; the subsequent report is integrated out. Classification only means that reported status changes but latent status does not. Genuine reversal means that latent status changes and returns to its original state in the next period; genuine persistence means that it remains in the new state. The three outcomes are mutually exclusive and exhaustive. The Table 1 columns use the nonstationary three-wave estimates; the final AR(2) column uses the four-wave Set 4 model with inconsistency, severity, and matching-quality predictors of symmetric classification error. Because the Table 1 and Table 5 estimation samples differ, comparisons combine model and sample differences.

Transition coefficients are on a common probit scale. The unrestricted AR(1) and no-duration AR(2) transition probabilities are re-expressed in their observationally equivalent probit parameterisations; the duration-controlled AR(2) is estimated directly by probit. Initial-state coefficients are logits, and the misclassification coefficient is $\text{logit}(2\pi)$ in all three original models and is reported directly for the staged extension. Parentheses contain analytical sandwich/delta-method standard errors. Duration variables and their missingness indicators vary across origin waves. The information-matrix condition numbers are 14.6 for AR(1), 130.7 for AR(2), 388,646 for duration-controlled AR(2), and approximately 746,558, 1,067,863, and 2,800,426 across the three Set 4 stages; every reported model is locally full rank.

Interpretation

- The AR(1) model estimates quarterly entry and exit of 3.20% and 3.13%, with a 2.70% symmetric misclassification probability.
- Duration controls substantially reduce the large AR(2) transition differences but do not eliminate them.
- With Set 4 controls and constant error, entry and exit after two periods in the same state are 4.72% and 4.56%; following a recent reversal they are 31.13% and 32.37%. Observed

transition heterogeneity therefore does not explain away the second-lag pattern.

- Allowing inconsistency and severity to predict error raises the posterior mean error rate from 0.93% to 2.02% and lowers the four transition rates to 4.22%, 27.52%, 28.65%, and 3.95%.
- Adding matching-quality indicators improves fit further and produces rates of 4.22%, 28.28%, 29.38%, and 3.94%, with 1.95% mean error. The substantive AR(2) pattern remains large after the complete observed-control sequence.
- Appendix Table A4C shows that the final model attributes 41.1% of apparent transitions to classification error, 16.8% to genuine transitions that reverse next period, and 42.1% to genuine persistent transitions. Among the 58.9% inferred to be genuine, 28.6% reverse in the following period. This is more balanced than the Table 1 symmetric-error model, which attributes 67.0% to error and only 1.0% to genuine short-lived transitions.
- Surviving AR(2) dependence may reflect unobserved duration, state dependence, latent heterogeneity, or within-quarter transitions; the current estimates do not distinguish these mechanisms.
- The extension is locally identified and has very small final scores, but its rising information condition number calls for sensitivity checks before it is treated as the preferred specification.

To do

- Compare AR(1) and AR(2) using formal fit criteria and richer duration controls.
- Profile the weakly curved directions in the three Set 4 extensions and check stability with additional long starts and alternative covariate scaling.
- Compare the staged extensions on a common sample with the simpler AR(1), AR(2), and duration-controlled benchmarks.
- With additional waves, test latent heterogeneity, wave-varying transitions, and correlated classification errors as competing explanations.

6. Unobserved heterogeneity: Table 6 and Appendix Table A5

- Re-estimated the two-type finite-mixture model by direct likelihood maximisation with multiple starts and audited local identification.
- Removed the underidentified three-wave FMM table: only the no-error model with free initial conditions is locally identified; symmetric error cannot be separated from two-type transition heterogeneity.
- Developed a four-wave alternative with observed covariates, duration controls, and class-specific misclassification intercepts.
- Compared all four specifications in one main table, including the nested no-misclassification model, and moved inconsistency-specific error rates to a secondary table.

Table 6. Comparison of all four-wave two-type FMM specifications

	No controls				Controls			
	No error		Common error		Common error		Type-specific error	
	Stable	Mobile	Stable	Mobile	Stable	Mobile	Stable	Mobile
Latent-class share (%)	75.65 (1.00)	24.35 (1.00)	92.13 (0.37)	7.87 (0.37)	89.63 (1.26)	10.37 (1.26)	58.17 (3.05)	41.83 (3.05)
Entry rate (%)	0.46 (0.18)	30.46 (0.75)	2.36 (0.07)	42.16 (1.15)	3.20 (0.12)	27.30 (2.49)	1.33 (0.15)	11.98 (0.64)
Exit rate (%)	0.004 (0.240)	33.63 (0.77)	2.21 (0.08)	47.04 (1.16)	3.01 (0.16)	26.87 (1.57)	0.75 (0.20)	11.18 (0.65)
Misclassification rate (%)	0.00 (fixed)	0.00 (fixed)	1.35 (0.03)	1.35 (0.03)	1.74 (0.03)	1.98 (0.03)	0.68 (0.09)	4.60 (0.18)
Initial employment rate (%)	47.73 (0.18)	48.17 (0.45)	47.74 (0.13)	47.95 (0.74)	47.38 (0.22)	50.59 (1.75)	44.61 (0.58)	52.33 (0.56)
Log-likelihood	-451,874.52		-450,805.93		-428,777.56		-428,568.60	
<i>N</i>	308,240		308,240		308,240		308,240	

All estimates except the log likelihood and observation count are percentages; analytical survey-weighted sandwich standard errors, transformed by the delta method, are in parentheses. Entry and exit rates are posterior risk-set weighted in the controlled models. Posterior mean misclassification averages each respondent-wave's fitted posterior mismatch probability using survey weights and posterior class membership. Controls comprise observed covariates and duration controls. Both controlled models use common slopes and class-specific transition intercepts. The last model additionally makes the misclassification intercept class-specific; the age- and education-inconsistency slopes remain common. In the no-error model, all misclassification probabilities are fixed at zero. The no-error model has full probability-Jacobian rank, but weak curvature around its near-zero stable-class exit estimate warrants caution. Pattern-specific implied misclassification rates are reported separately in Appendix Table A5.

Appendix Table A5. Misclassification rates by inconsistency pattern

Inconsistency pattern	Common error: Stable	Common error: Mobile	Type-specific error: Stable	Type-specific error: Mobile	Mobile minus stable
None	0.72 (0.02)	0.72 (0.02)	0.14 (0.04)	2.37 (0.15)	2.22
Age	14.72 (0.31)	14.72 (0.31)	6.28 (0.98)	35.52 (1.05)	29.25
Education	3.73 (0.12)	3.73 (0.12)	0.89 (0.21)	11.83 (0.53)	10.94
Age and education	34.85 (0.40)	34.85 (0.40)	23.62 (1.97)	46.93 (0.34)	23.31

Estimates are misclassification rates in percent; analytical standard errors are in parentheses. Both specifications include observed covariates and duration controls. The common-error model has a common misclassification intercept; the type-specific model has separate stable- and mobile-class intercepts. Age- and education-inconsistency slopes are common across classes.

Appendix Table A5, Panel B. Preferred FMM coefficients and diagnostics

Quantity	Estimate	Analytical SE
Persistence: log tenure	0.650	0.015
Entry: log time since work	-0.325	0.007
Entry: never worked	-0.860	0.018
Persistence: permanent contract	0.399	0.016
Persistence: informal sector	-0.266	0.016
Mobile-minus-stable misclassification intercept	2.839	0.225
Wald statistic for common error intercept	12.64	—
Entry-risk peak age	38	—
Persistence peak age	53	—
Information-matrix rank	32/32	—
Minimum information eigenvalue	0.0000120	—
Information condition number	29,216	—
Maximum weighted score	0.0000032	—

Coefficients are probit-index effects except the logit-scale difference between the class-specific misclassification intercepts. The controlled sample contains 308,240 people and collapses exactly to 229,809 likelihood cells.

Interpretation

- The three-wave symmetric-error FMM is underidentified and should not be reported structurally; four waves are required to separate the two mechanisms.
- Imposing no misclassification pushes the stable-class exit estimate to essentially zero and attributes 30–34% quarterly transition rates to the mobile class. Allowing error materially improves fit and avoids this extreme decomposition.
- Four waves support a large stable/mobile distinction after controlling for observed characteristics and duration.
- Allowing the misclassification probability to differ by type makes the mobile class larger and less extreme, showing that the uncontrolled FMM mixed genuine mobility, observed heterogeneity, and heterogeneous classification error.
- In the preferred controlled model, the survey-weighted posterior mean misclassification rate is 0.68% in the stable class and 4.60% in the mobile class, a much sharper reliability contrast than the baseline probabilities alone convey.
- The type-specific probabilities describe joint latent classifications; they do not show that mobility causes reporting or matching error.
- The four-wave results are best presented as a replacement or appendix extension rather than as validation of the underidentified three-wave results.

To do

- Complete profile-likelihood, multistart, and simulation checks for the four-wave model’s identification and class separation.
- Test covariate-dependent class membership and, with a longer panel, whether latent types can be combined with AR(2) state dependence.

7. Endogenous duration dependence: Table 7

- Modelled transition hazards as functions of endogenous tenure and time since work, separately from the age/education-consistency analysis in Table 3.
- Repaired the tenure-contamination likelihood and optimisation diagnostics.
- Retained exact consistency between clean duration reports and latent employment histories.
- Integrated unavailable wrong-state duration clocks instead of imputing them.
- Replaced constant and power-law transition hazards with tail-safe piecewise hazards.
- Added contamination in reported time since work and tested local and joint alternatives.

Table 7: Tenure-contamination misclassification and transition rates

	Constant, exact	Power law, exact	Piecewise, exact	Independent error	Local error	Joint per-wave error
Entry rate	4.08 (0.01)	1.96 (0.01)	3.64 (0.04)	3.13 (0.03)	2.91 (0.03)	3.15 (0.03)
Exit rate	4.26 (0.01)	4.51 (0.01)	4.62 (0.01)	4.42 (0.01)	4.43 (0.01)	4.42 (0.01)
Misclassification rate	10.21 (0.03)	10.53 (0.03)	9.93 (0.03)	2.33 (0.02)	3.02 (0.02)	2.32 (0.02)
Initial employment rate	54.98 (0.08)	55.92 (0.08)	55.02 (0.09)	48.30 (0.08)	48.75 (0.08)	48.30 (0.08)
Tenure contamination rate	24.79 (0.07)	25.09 (0.07)	24.98 (0.07)	25.22 (0.06)	25.22 (0.06)	25.21 (0.06)
Timegap contamination rate	0.00 fixed	0.00 fixed	0.00 fixed	28.04 (0.11)	39.20 (0.15)	16.62 (0.07)
Log likelihood	-2,510,430.3	-2,475,676.8	-2,462,197.4	-2,330,075.3	-2,343,051.4	-2,320,294.9
N	402,923	402,923	402,923	402,923	402,923	402,923
Transition-hazard form	Constant	Power law	Piecewise	Piecewise	Piecewise	Piecewise
Tenure-error model	Independent marginal	Independent marginal	Independent marginal	Independent marginal	Independent marginal	Independent marginal
Timegap-error model	None (exact)	None (exact)	None (exact)	Independent marginal	Local category	Joint per-wave marginal

Notes. Rates are in percent and follow the common entry, exit, misclassification, and initial-employment order used in the other main tables; model-specific duration-error rates follow. Entry and exit are posterior risk-set and survey-weighted quarterly probabilities. Parentheses contain numerical observed-information/delta-method standard errors from the full covariance matrix. In every model, a contaminated tenure report is an independent draw from the marginal tenure distribution; otherwise it must follow the exact clock. ‘Exact’ means that no separate timegap-contamination process is allowed; status misclassification and tenure contamination remain allowed.

For the joint model, 16.62% is the per-report probability. It implies a 30.48% probability that at least one report in a two-report pair is contaminated.

Appendix Table A6, Panel A. Exact and alternative duration profiles

Duration interval	Exact exit hazard	Exact exit (%)	Independent- error exit (%)	Joint-model exit (%)
0–3 months	0.3420	8.35	7.79	7.79
3–12 months	0.3558	8.51	7.90	7.90
1–3 years	0.1757	4.30	4.23	4.23
3–5 years	0.1522	3.73	3.67	3.67
5+ years	0.1353	3.33	3.27	3.27

Duration interval	Exact entry hazard	Exact entry (%)	Independent- error entry (%)	Joint-model entry (%)
0–3 months	0.5060	9.15	7.66	7.88
3–12 months	0.2620	6.34	5.00	5.07
1–3 years	0.1098	2.71	2.27	2.27
3–5 years	0.1177	2.90	2.57	2.57
5+ years	0.1425	3.50	2.82	2.83

Appendix Table A6, Panel B. Auxiliary model comparisons

Quantity	Constant, exact	Power law, exact	Piecewise, exact
Parameters	5	7	13
AIC	5,020,870.7	4,951,367.6	4,924,420.8
Mean employment duration (months)	68.9	68.1	67.0
Mean nonemployment duration (months)	72.0	1,257.2	76.7
Comparable two-report timegap contamination (%)	0	0	0
Analytical SE	fixed	fixed	fixed

Quantity	Independent error	Local error	Joint per-wave error
Parameters	14	14	14
AIC	4,660,178.6	4,686,130.8	4,640,617.9
Mean employment duration (months)	69.6	69.6	69.6
Mean nonemployment duration (months)	95.6	102.2	94.8
Comparable two-report timegap contamination (%)	28.04	39.20	30.48
Analytical SE	0.11	0.15	0.12

Appendix Table A6, Panel C. Effect of integrating unavailable wrong-state clocks

Quantity	Imputed clock	Integrated clock
Entry rate (%)	2.08	1.96
Exit rate (%)	4.91	4.51
Misclassification rate (%)	10.35	10.53
Tenure contamination rate (%)	24.81	25.09
Exit duration slope	-0.4413	-0.3742
Entry duration slope	-0.9148	-0.8782
Mean nonemployment duration (months)	3,800	1,257

Appendix Table A6, Panel D. Timegap-contamination mechanism decomposition

	Clock-feasible	Reset-compatible	Other	Posterior
Interview interval	(%)	contradiction (%)	contradiction (%)	contamination (%)
Wave 1 to wave 2	84.42	2.82	12.76	28.20
Wave 2 to wave 3	84.95	2.98	12.07	27.87

In the power-law model, exit falls from 7.42% at zero tenure to 3.22% at ten years, while entry falls from 8.51% to 1.18%. Integrating rather than imputing wrong-state clocks affects 15.5% of the

posterior employment risk set and 4.5% of the nonemployment risk set. In the piecewise exact model, median employment and nonemployment durations are 35.3 and 51.4 months. For the independent-error model, clock-impossible shares are 15.6% and 15.0%, while average contaminated-branch probabilities among feasible reports are 14.96% and 15.09%. Roughly 10% of fitted contamination mass is reset-compatible, 43–45% comes from contradictions that a hidden reset cannot explain, and 45–46% is allocated among clock-feasible reports. Joint-model exit risks differ from the independent-error risks by less than 0.03 percentage points in every interval.

The independent-error information matrix has rank 14/14, condition number 45, and a sharply peaked likelihood around 28.04%; the boundary likelihood-ratio diagnostic is 264,244. Refined maximum scaled gradients are 2.0×10^{-4} for the exact piecewise fit, 2.1×10^{-6} for the local-error fit, and 2.5×10^{-6} for the joint model. These are numerical diagnostics, not substitutes for the standard errors reported in Table 7.

Interpretation

- Piecewise hazards fit much better than constant or power-law hazards and improve the weighted-likelihood AIC by about 26,947 relative to the power law, while avoiding its implausibly long nonemployment tail.
- Exit falls after the first year of tenure. Entry falls sharply with time since work and then levels off rather than approaching zero.
- Allowing timegap contamination reduces estimated status error from about 10% to 2–3%, while tenure contamination remains close to 25%.
- The joint per-wave model fits best and leaves transition rates and duration profiles almost unchanged relative to the simpler contamination model.
- Only about 3 percentage points of timegap contradictions can be explained by an unobserved within-quarter employment spell; hidden resets are not the main source of the contamination estimate.
- The results identify inconsistency among status, tenure, timegap, and linked histories. They do not determine whether the underlying cause is respondent error, matching error, or another data problem.

To do

- Add bootstrap or profile uncertainty and posterior-predictive checks for the joint status and duration distributions.
- Test alternative contamination densities, rounding and heaping rules, and tail cut-points while retaining transition-duration consistency.

8. No-error duration counterfactual (Appendix Table A7)

- Estimated flexible AR(1) and AR(2) transition models treating reported status changes as error-free.
- Simulated the duration clocks implied by those transitions.
- Compared simulated and observed tenure and timegap distributions, with transition-conditional graphs as the primary diagnostic.

Appendix Table A7: No-error duration-clock contradictions

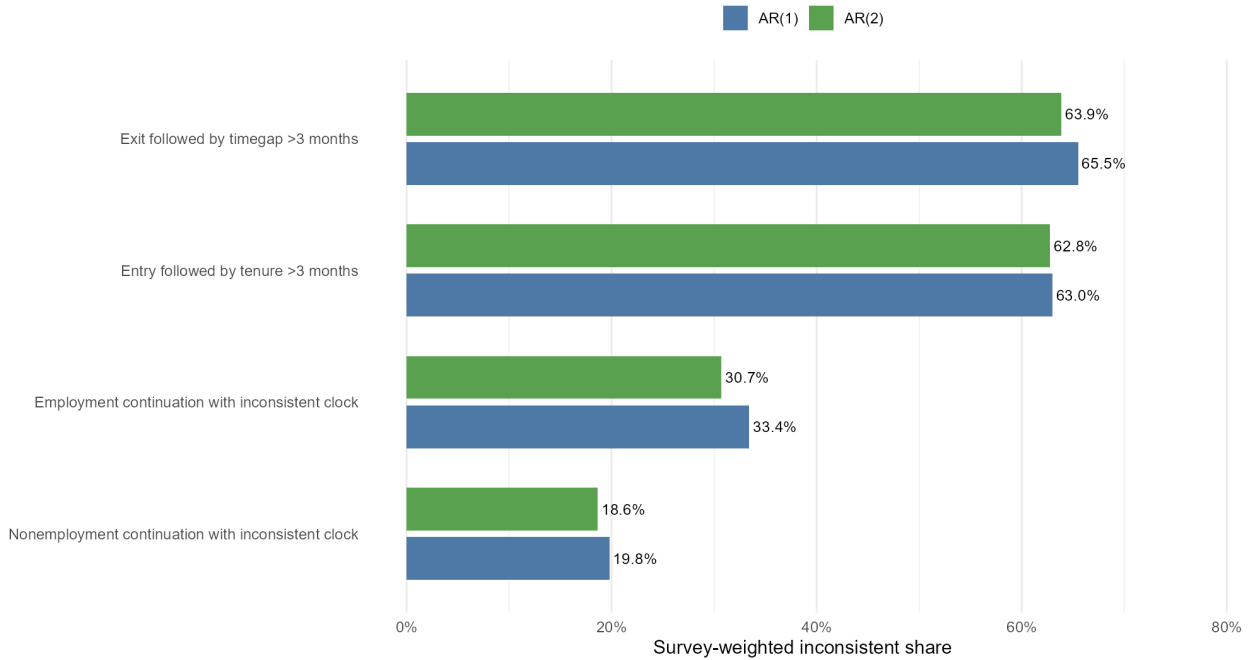
	AR(1)	AR(2)
Apparent entry followed by tenure over three months	63.0 (0.30)	62.8 (0.35)
Apparent exit followed by timegap over three months	65.5 (0.29)	63.9 (0.35)
Employment continuation with inconsistent tenure	33.4 (0.11)	30.7 (0.13)
Nonemployment continuation with unreachable timegap	19.8 (0.09)	18.6 (0.10)

Notes. Entries are survey-weighted shares in percent. Parentheses contain person-clustered linearized standard errors. Each no-error model allows flexible duration dependence in observed transitions; destination-wave durations are held out and used for these clock-consistency diagnostics.

Graph accompanying Panel A. Clock contradictions under AR(1) and AR(2)

Share of reported histories that violate a no-error spell clock

One-month tolerance is allowed for tenure increments; timegap uses reported category intervals



Appendix Table A7, Panel B. Observed transition benchmarks

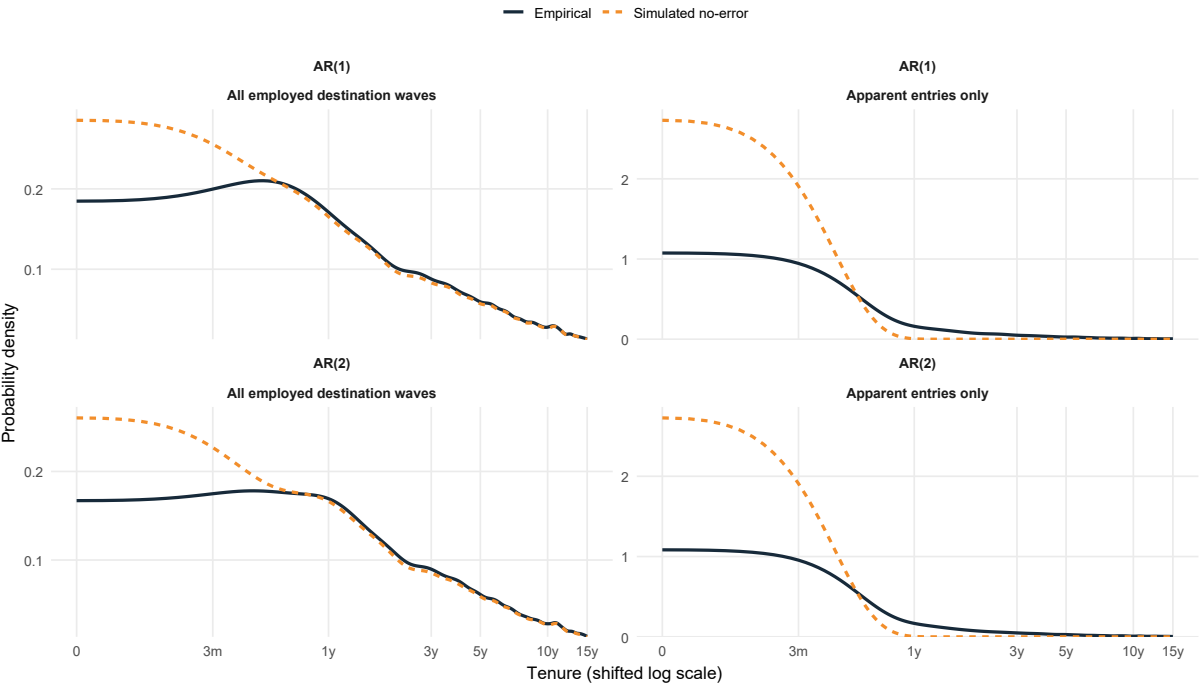
Rate	AR(1) (%)	AR(2) (%)
Entry rate	8.23	7.78
Exit rate	8.47	7.83
Admissible tenure after apparent entry	37.0	37.2

The flexible transition equations reproduce these observed entry and exit rates. Among apparent exits in the AR(1) analysis, 26.0% report that they have never worked, reinforcing the contradiction

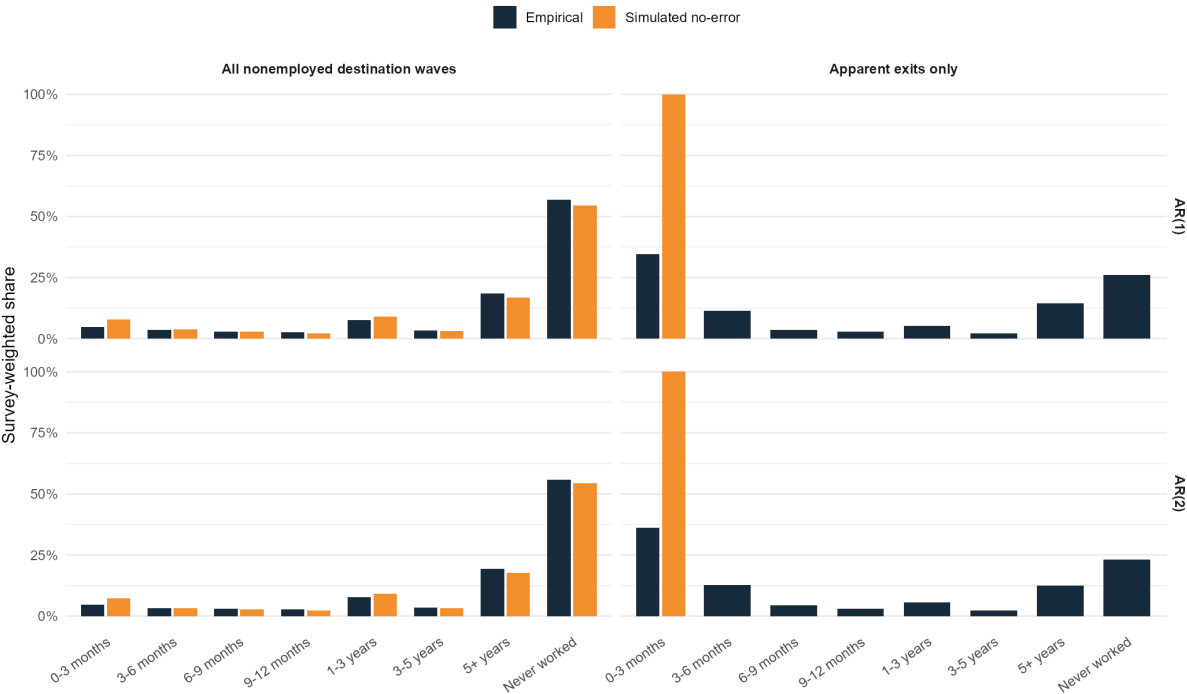
between status changes and the reported duration history.

Graphs accompanying Panel B. Empirical and simulated duration distributions

Empirical and simulated employment-tenure densities
Survey-weighted densities; log(tenure + one month) axis and common three-month bandwidth



Empirical and simulated time-since-employment distributions
Reported categories are compared with clocks propagated or reset by the no-error model



Interpretation

- Flexible no-error transition models reproduce average observed changes but cannot reconcile them with destination-wave duration reports.
- Most apparent entries and exits imply duration clocks that are impossible under a genuine transition during the interview interval.
- Marginal duration distributions obscure the problem because most respondents do not change reported state; transition-conditional comparisons are more persuasive.
- The diagnostic motivates jointly modelling status and duration consistency without claiming which reported variable or linked record is incorrect.

To do

- Add simulation uncertainty bands to the empirical-versus-implied duration graphs and report transition-conditional comparisons prominently.
- Repeat the counterfactual under alternative matching panels and initial-clock assumptions.

Reproducibility and remaining inference

- A result-by-result audit against `branch_changes_summary.Rmd` maps the baseline estimates to Table 1/A1, matching-rule results to Table 2, age/education-consistency results to Table 3/A2, covariate estimates and hazard distributions to Table 4/A3, AR(2) results to Table 5/A4, four-wave FMM results to Table 6/A5, tenure-contamination results to Table 7/A6, and the no-error duration benchmarks to A7. No numerical estimate reported in the long summary is intentionally omitted here; only detailed implementation narrative is abbreviated.
- Tables 1, 3, and 4 and Appendix Tables A1–A3 are rebuilt by `build_tables.R`.
- Table 2 is rebuilt by the matching-sensitivity script; the four-wave, tenure, and diagnostic scripts save the inputs for Tables 5–7 and Appendix Tables A4–A7 outside version control.
- Analytical standard errors are available for Tables 1–7 and their associated model estimates. Table 7 uses the full numerical observed-information matrix and delta method; Appendix Table A7 uses person-clustered linearization for its weighted descriptive shares.
- Table 7’s numerical observed-information errors should still be checked with a person-level bootstrap or profile likelihood before final submission.
- A person-level bootstrap remains the most useful common inference robustness check across the paper.

Detailed changes to code and estimation

The preceding numbered sections, tables, interpretations, and to-do lists are identical to the short summary. This section records the additional estimation and implementation detail needed to reproduce and audit those results.

Tables 1–2: exact baseline likelihood and matching robustness

The three-wave baseline estimator works with the exact eight possible observed employment histories. Collapsing to these histories preserves the weighted observed likelihood while making multiple-start optimization inexpensive. All probabilities are represented by unconstrained logits. Stationary models impose the restriction inside the likelihood; free-initial-condition models estimate the initial employment probability separately.

The pipeline uses nested warm starts and dispersed deterministic starts, checks the analytical score against numerical derivatives, requires positive-definite observed information, and verifies that each unrestricted specification weakly dominates its nested restriction. Saved fits carry the panel identity, sample signature, sample size, total weight, likelihood, and optimizer diagnostics so the table builder can reject stale results.

Table 2 reruns the nonstationary no-error and symmetric-error models on panels A, B, and C. These panels apply progressively stricter longitudinal matching rules. The sensitivity script changes the panel input explicitly rather than silently modifying the default ingestion path, which remains `df_qlfs_A.rds`.

Table 3: observed indicators of linked-record reliability

The inconsistency likelihood permits the symmetric status-error probability to depend on age inconsistency, education inconsistency, both inconsistencies, and in the expanded models the magnitude of those discrepancies. The most recent specifications also include indicators for observations retained in panel B but not C and in panel A but not B. These indicators are constructed using the same nested matching algorithm as Table 2.

The EM update for the error equation now maximizes its correct conditional objective. Both stationary and free-initial-condition results are retained, and the robustness model allows true entry and exit rates to differ across reliability groups. Reported means use the same survey and posterior weighting conventions as the other tables. The mechanism language remains deliberately neutral: inconsistencies may reflect response problems, incorrect longitudinal matches, or both.

Table 4: observable transition heterogeneity

Sets 1–4 are nested. Set 1 contains age, age squared, and education; Set 2 adds race and sex; Set 3 adds origin-wave log tenure to persistence and origin-wave log time since work plus never-worked status to entry; Set 4 adds origin-wave contract type and informal-sector status to persistence. Contract type and sector may change between transitions. They do not enter entry because they are undefined for a genuinely nonemployed origin state, so respondents are retained without assigning artificial job characteristics.

The eight constant-error/no-error specifications retain the repaired generalized-EM implementation: normalized weights, joint transition-block updates, likelihood-preserving backtracking, nested starts, and explicit convergence and nesting checks. Posterior risk-set and survey-weighted hazards

replace unweighted averages. Individual-level survey-weighted sandwich covariances and delta-method transformations provide approximate analytical errors where bootstrap estimates are unavailable.

Column (9) is estimated by the exact observed likelihood rather than by the old separable M-step. It combines the Set 4 transition equations with a symmetric error probability depending on the full Table 3 reliability specification: age and education inconsistencies, severity indicators, panel B but not C, and panel A but not B. Five starts were evaluated. The polished solution has 30 of 30 information-matrix directions identified, condition number about 18,126, and maximum normalized score 1.1×10^{-8} . The new mean entry, exit, and error rates are 4.67%, 4.62%, and 2.55%. Detailed error slopes and implied reliability-specific rates appear in Appendix Table A3B.

Table 5: four-wave AR(1) and AR(2)

The four-wave modules enumerate latent histories and evaluate the exact collapsed observed likelihood. The AR(1) benchmark has free initial employment and symmetric error. The unrestricted AR(2) allows the previous two latent states to determine the next transition. Its duration-controlled specification adds the same state-appropriate tenure and time-since-work variables used in Table 4 and permits those variables to change with the origin wave.

The analytical covariance uses the observed information and collapsed person-level score outer products. The duration-controlled AR(2) information matrix is locally full rank, although its high condition number means the initial-state and missing-duration directions warrant caution.

The new extension is estimated as three exact nested likelihoods on the same 308,240 observations. The first combines AR(2) with the Table 4 Set 4 transition equations and constant symmetric error. The second adds age/education inconsistency and severity predictors to the error equation. The third adds panel B-not-C and panel A-not-B matching-quality predictors. Stage-specific collapsing leaves 197,933, 220,795, and 228,195 exact cells. Each stage uses two substantively distinct starts, checks likelihood nesting, applies an observed-information Newton correction where useful, and is accepted only with a maximum normalized score below 2×10^{-5} and full information rank. The final scores are 2.8×10^{-6} , 4.2×10^{-6} , and 2.8×10^{-6} . All three models are locally full rank, but condition numbers rise from about 746,558 to 2,800,426, motivating profile and long-start sensitivity checks. Appendix Table A4C is generated from a separate prospective decomposition that conditions only on the two reports forming an apparent transition and integrates out the subsequent report.

Table 6: finite-mixture heterogeneity

The two-type four-wave model uses exact likelihood refinement after EM warm starts, label normalization, multiple starts, and analytical sandwich/delta inference. The comparison now contains the no-error FMM, the uncontrolled symmetric-error FMM, the controlled common-error model, and the controlled type-specific-error model in one table. Observed transition slopes are common across latent types in the parsimonious controlled models, while type-specific intercepts retain mobility heterogeneity.

Age and education inconsistencies determine error risk without assigning them to a particular behavioral mechanism. The preferred type-specific model allows the two classes to have different error intercepts and common inconsistency slopes. Its local information matrix is full rank. The failed three-wave symmetric-error FMM is not reported as an identified result: three waves do not separate two-type transition heterogeneity from symmetric misclassification.

Table 7: endogenous duration dependence

The tenure-contamination module preserves the central consistency restriction: latent employment paths must agree with clean reported tenure and time-since-work clocks. Successive specifications relax constant transition hazards, integrate unavailable wrong-state clocks, use tail-safe piecewise hazards, and allow alternative contamination processes for tenure and timegap reports. These are likelihood changes, not presentation-only adjustments.

The preferred timegap treatment is the joint per-wave marginal-contamination model. The table reports approximate analytical errors from the full numerical observed-information matrix and delta method, not bootstrap errors. Appendix Table A6 retains hazard profiles, auxiliary comparisons, wrong-state-clock integration checks, and the within-quarter reset decomposition.

Appendix Table A7: no-error duration counterfactual

The counterfactual diagnostic estimates flexible no-status-error AR(1) and AR(2) transition models without using destination-wave durations in the likelihood. It then propagates implied spell clocks and compares them with the held-out reported tenure and timegap. Large conditional contradictions show that reported employment changes and duration histories cannot generally both be correct. The exercise motivates joint modeling but does not distinguish response inconsistency from longitudinal matching error.

Reproducibility and generated files

`build_tables.R` generates Tables 1, 3, and 4 and their coefficient appendices; the matching-sensitivity runner generates Table 2. The four-wave and duration modules retain their own reproducible runners, tests, result summaries, and diagnostic outputs. Confidential data and intermediate fitted objects remain ignored by Git. Curated paper-facing snapshots and both summary sources are tracked. A complete rerun still requires the confidential QLFS inputs and the R environment recorded by `renv`.

The remaining inference priorities are the person-level bootstrap for the main covariate and duration specifications, robustness to alternative reliability links, and likelihood-profile and longer-start checks for the ill-conditioned Table 5 Set 4 AR(2) extensions.